Small Fine-Tuned Planners with Execution-Verified Data and Calibrated Abstention for Tabular Canonicalization

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Abstract

Cleaning messy tabular data—particularly *canonicalization*, the merging of inconsistent surface forms such as `USA`/`U.S.A`/`united states` into one canonical value—resists rule-based automation and is routinely done by hand. We present ScrubData, a system in which a small ($\leq$4B-parameter) language model acts as a *planner*: it reads an aggregated column profile (per-value frequency counts, invariant to row count) and emits a structured JSON cleaning plan that a deterministic executor applies, making every change auditable and reversible. Three design decisions carry the result. (1) *Execution-verified training data*: every synthetic training example is kept only if executing its plan provably recovers the known-clean table. (2) *Reference grounding*: canonical forms are never free-generated; values are reconciled against type-scoped reference taxonomies (GeoNames, ISO) with fuzzy retrieval. (3) *Calibrated abstention*: below a confidence threshold the system declines to act and flags values for review; we show the confidence is selective-prediction-grade (AURC 0.120, precision rises monotonically with threshold). On alias-level canonicalization, our fine-tuned 4B planner reaches micro-F1 0.901 versus 0.452 for a much larger zero-shot model and 0.152 for a rule heuristic. On a 65-dataset validation suite with a churn-neutral metric, grounding with abstention matches frequency-based clustering on real-error F1 while perfectly rejecting adversarial trap values that clustering systems wrong-merge. The same architecture generalizes to PII protection: a 44M token classifier types columns from bare cell values (100% on names/addresses), and deterministic masking yields zero residual detectable PII. All components run locally on commodity hardware; no data leaves the machine.

1 Introduction

A large share of practical data work is cleaning: a sales export where the same country is spelled four ways, a hospital roster where `birminghxm` should be `birmingham`, a CRM dump with mixed date formats and duplicated contacts. The fuzzy half of this work—recognizing that distinct surface forms denote the same entity—is exactly what rules do poorly and humans do slowly.

Large language models can do this fuzzy matching, but deploying them as cell editors has three problems. First, *trust*: a model that edits cells directly can silently corrupt data, and its errors are unauditable. Second, *cost and privacy*: shipping every row of a private table to a hosted frontier model is expensive and often unacceptable. Third, *hallucination*: asked for a canonical form, a generative model will invent one, and on tail entities it will invent wrong ones.

ScrubData addresses all three with an architecture in which the model never touches data. A profiler aggregates each column into a value-frequency distribution; a small local model reads the profile and *proposes* a JSON cleaning plan; a deterministic pandas executor *applies* it. The plan is the complete, inspectable, reversible specification of every change—there are no silent edits by construction (§3). Because the prompt scales with the number of *distinct* values rather than rows, a million-row table profiles like a hundred-row one.

This paper makes four contributions:

1. **A planner/executor decomposition for data cleaning** with an auditable trust contract: the model proposes, a deterministic engine executes, every decision is logged with lineage (§3).

2. **Execution-verified synthetic supervision**: a data-generation method in which every training example is validated by running the executor on the (dirty table, plan) pair and checking that the known-clean table is recovered; non-recovering examples are discarded (§3.2).

3. **Reference-grounded canonicalization with calibrated abstention**: canonical forms come from type-scoped reference taxonomies via fuzzy retrieval with an ambiguity margin; below-threshold matches abstain and surface for review. We characterize the confidence with risk–coverage analysis (§3.3, §5.4).

4. **An un-gameable evaluation**: a 65-dataset suite (real-error benchmarks plus seeded error injection over 20 harvested open-data domains) scored with a churn-neutral, convention-tolerant metric that cannot be inflated by mass rewriting; we report two metric artifacts that our own ablations exposed and corrected (§4).

We deliberately report a negative-flavored finding alongside the positive ones: on *injected* typos, classical frequency clustering remains a strong baseline—by construction, injection places the canonical form in the column, which is clustering's ideal regime. The advantage of grounding is concentrated where it matters: real errors, tail entities absent from the column, and adversarial near-misses where acting at all is wrong (§5).

2 Related Work

Error detection and repair. Raha and Baran [1] established configuration-free error detection and correction benchmarks (hospital, beers, flights, rayyan), which we adopt as out-of-distribution evaluation. HoloClean [2] combines integrity constraints, external reference data, and statistics in probabilistic repair, demonstrating that external signals can veto statistically plausible but wrong repairs—an insight our reference-veto inherits. GARF [3] learns repair rules self-supervised from the data itself; it also demonstrates the structural limit we observe for frequency-only methods: a lone categorical column offers no co-occurring signal to vote against an error.

LLMs for data wrangling. Narayan et al. [4] showed frontier foundation models handle entity matching and imputation few-shot; Jellyfish [5] and Table-GPT [6] fine-tune mid-size models for data tasks. RetClean [7] is closest in spirit: retrieval from data lakes grounds cell repair, with the key empirical split that parametric knowledge suffices on world-known head values but collapses on the tail—motivating retrieval. Our work differs in the planner/executor decomposition (the model emits no cell values, only plans), in execution-verified supervision, and in the calibrated-abstention contract.

Entity linking over tables. TURL [8] and TableLlama [9] inject candidate entities into table understanding; Belotti et al. [10] show retriever coverage is the accuracy ceiling for table entity disambiguation and that long candidate lists hurt smaller models. RACOON [11] shows inference-time KG retrieval lifts a frozen model substantially, supporting our choice to ground at inference rather than bake aliases into weights (TURL's out-of-domain collapse is the cautionary result). MTab [12] established type-constrained matching with abstention in semantic table annotation.

Clustering-based cleaning tools. OpenRefine's key-collision (fingerprint) and nearest-neighbour clustering are the de-facto practitioner baseline; we reimplement both faithfully (including blocking) and compare head-to-head.

Selective prediction. Risk–coverage analysis and calibration metrics [13] formalize "knowing when not to act"; to our knowledge their application to data-cleaning merge decisions is new.

Small specialized models. OpenMed [14] fine-tunes sub-500M encoders to state-of-the-art biomedical NER, the sister result to our thesis that small specialized models beat large generic ones on narrow structured tasks; we adopt their released PII token classifiers for column typing (§3.4).

3 Method

3.1 Planner / executor decomposition

A *profiler* reduces each column to a typed summary: detected semantic type, missing counts, issue flags, and a value–frequency distribution capped at 80 distinct values (high-cardinality columns are summarized by their head). The *planner*—either a deterministic heuristic or our fine-tuned 4B model—maps the profile (plus three sample rows) to a JSON plan: a list of per-column operations drawn from a closed vocabulary (`canonicalize_categories` with an explicit mapping, `parse_date`, `standardize_phone`, `mask_pii`, . . .), table operations, and review flags. The *executor* applies the plan with pure pandas transforms. The plan is the only channel through which data changes: every diff is attributable to a named operation with a rationale, the original table is never mutated, and abstentions are first-class plan objects. We export per-run decision summaries as OpenTelemetry GenAI spans.

3.2 Execution-verified synthetic supervision

Training pairs are generated by corrupting clean synthetic tables with realistic noise (casing, aliases, single-character typos with Zipf-distributed long-tail categorical columns of 30–80 distinct values) while recording the ground-truth plan. The defining step is *verification by execution*: a candidate example is kept only if $\textsc{Execute}(\text{dirty}, \text{plan}) = \text{clean}$ cell-for-cell. This closes the loop between supervision and semantics—a plan that would not actually clean the table can never become a training label. We augment with real supervision derived from paired dirty/clean benchmarks by aligning cells and keeping only *learnable* canonicalizations (a surface form that is a string variant of its target and never a legitimate value elsewhere), which excludes unlearnable per-cell corrections such as divergent flight times. The fine-tune is QLoRA (rank 32) over Qwen3-4B-Instruct in bf16; one practical finding is that the base model's tool-calling prior dominates free-running generation even after convergent fine-tuning (loss 0.16) and must be suppressed at decode time by banning the two tool-call tokens.

3.3 Reference-grounded canonicalization with abstention

For columns whose values reconcile to a known concept type (countries, administrative regions, cities), canonical forms are never generated: a fuzzy retriever (normalized edit similarity with first-character blocking and length prefilters) matches each distinct value against the type-scoped reference (ISO/pycountry; GeoNames cities500, 196k entries). A value maps to a canonical only if (i) similarity clears a threshold τ=0.84, (ii) the best–second-best margin clears 0.03 (ambiguity veto: a value equally close to `Box` and `Boaz` abstains), and (iii) the canonical is cast to the column's observed case convention. Near-misses ($0.70 \le s < \tau$) are surfaced as review flags. The same wrapper grounds the *model* planner: for reference-typed columns the model's free-generated mapping is replaced by the grounded one, so the model can add coverage but never invent a canonical for a grounded type.

3.4 PII as a second task instance

The identical contract covers PII: a deterministic tier types columns by checksum and pattern validators (Luhn, IBAN mod-97, SSN/email/phone) over distinct values; an optional 44M OpenMed-PII token classifier extends coverage to names and addresses, gated by a sensitive-type allowlist and a column-level coverage vote. High-sensitivity checksum-confirmed columns are masked format-preservingly (keep-last-four), contact columns are flagged for review, and identifier columns are excluded from numeric reformatting. Masking, salted hashing, and join-stable pseudonymization are deterministic executor operations.

4 Evaluation Design

Suite. Five real-error benchmarks (Raha) plus seeded error injection (typo/OCR/case/whitespace) over 15 harvested open-data domains (NYC, Chicago, SF, LA, Seattle, Texas, WA portals; GitHub) $\approx$ 65 datasets per seed. We aggregate as a *double macro*—mean over error types of mean over datasets, harmonically combined with the domain macro—so no single table or error type dominates.

Churn-neutral metric. A cell change that is case/whitespace-equivalent to the input but does not restore the gold counts as nothing: not a fix, not a change, not damage. Without this, mass case-rewriting inflates precision (we observed +0.12 NORTH from *removing* case matching before the correction); with it, fixing a case-injected error requires actually acting. We additionally report *damage*—the rate of semantically corrupting clean cells—and an adversarial *abstain slice* whose traps are garbage strings (not single-edit variants of any reference entity; an earlier trap set mis-scored grounding for correctly mapping `Boazz`→`Boaz`). We report both repairs of these metric artifacts as evidence that gameability must be tested, not assumed.

Real vs. injected. Injected typos are in-distribution for frequency clustering by construction (the canonical is present and dominant in the column), so we report the real-error and injected slices separately.

5 Results

5.1 Small fine-tuned planner vs. large generic model

On frozen synthetic gold, the fine-tuned 4B planner reaches canonicalization micro-F1 0.901 (operation-F1 0.957, JSON validity 0.950), versus 0.452 for a much larger zero-shot generalist prompted identically and 0.152 for the rule heuristic. Multi-seed: TBD. On real hospital typos the synthetic-only fine-tune scores 0.000 repair recall; adding 20% real-derived supervision lifts it to 0.424—real data is what transfers.

5.2 Grounding vs. clustering

On hospital's real errors, grounded reconciliation beats frequency clustering on recall (0.257 vs 0.193), nearly doubles precision (0.053 vs 0.028), and cuts wrong changes by 62%. On the full suite (Table 1), grounded cleaning is statistically tied with clustering on real-error F1 (TBD

Table 1: Wide-suite comparison, 3 injection seeds, churn-neutral metric. NORTH is the double-macro harmonic mean; REAL-F1 is the real-error slice. (Filled from the final run.)

System	NORTH	±95%CI	REAL-F1	damage	abstain
Grounded (ours)	TBD	TBD	TBD	TBD	1.000
OpenRefine fingerprint	TBD	–	–	–	–
OpenRefine kNN	TBD	–	TBD	TBD	–
Grounded 4B model	–	–	TBD	–	TBD

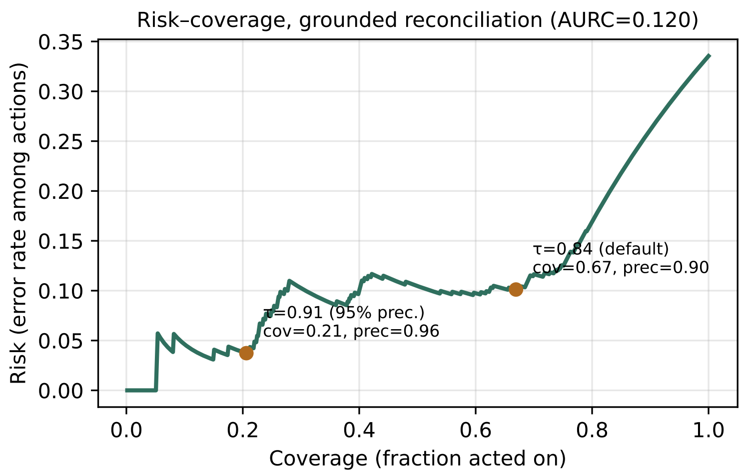

Figure 1: Risk–coverage for grounded city reconciliation (650 probes). Operating points annotated; the confidence supports thresholded abstention.

vs 0.135) while clustering wins the injected slice—its home regime. The behavioral difference is where grounding earns its place: perfect abstention on adversarial traps (1.000) versus 0.250 when abstention is removed, and reference-vetoed wrong merges that clustering commits (e.g. `guntxrsvillx`→`huntsville`).

5.3 Ablations

Removing abstention costs −0.038 NORTH, raises damage to 0.139 (from 0.118), and collapses trap abstention to 0.250. Removing the ambiguity margin costs −0.018 with +0.007 damage. Removing case matching costs −0.008 under the churn-neutral metric (and *gained* +0.12 under the uncorrected metric—the artifact). Replacing grounding with frequency clustering gains +0.049 NORTH, all of it from the injected slice (§4); real-error F1 is unchanged within noise.

5.4 Calibration of abstention

On a probe of reference-entity typos plus garbage traps, retrieval confidence is a usable selective-prediction signal: AURC 0.120, ECE 0.169 (mildly over-confident), and precision rises monotonically with threshold—0.899 precision at the default τ=0.84 (coverage 0.669), and ≥95% precision at τ=0.91 (coverage 0.206). Figure 1 shows the risk–coverage curve.

5.5 PII transfer and detection

The 44M OpenMed-PII classifier, trained on sentence-level clinical text, transfers to bare cell values without templates: 100% detection on person-name cells and 100% on addresses. Raw entity hits on non-PII columns (43%) are quasi-identifier typings (`city`, `occupation`) eliminated by the sensitive-type allowlist and coverage vote. The deterministic validator tier, evaluated out-of-distribution on per-type columns built from the Gretel PII test split (chosen specifically because OpenMed trains on Nemotron-PII), types **`5/5`** covered PII types correctly with **`0/7`** false positives on negative columns drawn from real open data. After deterministic masking, re-running all validators over the output finds zero (0/360) residual detectable PII.

6 Limitations

Reference coverage is the recall ceiling: entities absent from the taxonomy abstain by design, which is safe but not helpful; coverage work (larger gazetteers, ROR for organizations) moves recall directly. Our damage metric is convention-tolerant for case and whitespace but still counts alias expansion (`NYC`→`New York`) as damage when the gold keeps the alias—a value-level convention question we leave open. The confidence signal is mildly over-confident (ECE 0.169); temperature scaling is future work. The injected half of the suite, while seeded and reproducible, inherits the injector's error model; we mitigate with the real-error slice and report both. PII coverage is English-only, and we make no de-identification guarantee. Finally, the fine-tune headline is reported with multi-seed confidence intervals, but the wide-suite model row is single-seed for cost reasons and scoped as such.

7 Conclusion

A small, locally-run planner with execution-verified supervision, reference grounding, and calibrated abstention turns LLM data cleaning from a trust liability into an auditable system: every change is a named, reversible operation; uncertain actions become review flags rather than silent corruptions; and the evaluation itself is built to resist gaming. We believe the recipe—propose/execute decomposition, verification-by-execution, retrieval-grounded outputs, and selective prediction—is a template for deploying small specialized models on other structured tasks.

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